Unit 7 Pre-Test

Scatterplots & Two-Way Tables

Standards assessed: NC.8.SP.1–SP.4

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.SP.1 · NC.8.SP.2 Describe patterns of association in bivariate data.

• *I can describe associations in scatterplots and name their key features.*

Match each vocabulary term to its definition, then answer each question.

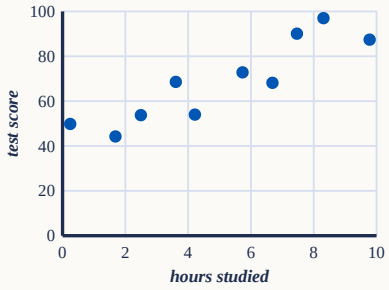
C scatterplot ✓	**A.** As one variable increases, the other tends to increase.
A positive association ✓	**B.** A data point that lies far away from the rest of the data.
D negative association ✓	**C.** A graph that plots paired data as points.
B outlier ✓	**D.** As one variable increases, the other tends to decrease.
E line of best fit ✓	**E.** A straight line drawn to model the trend in a scatterplot.
F cluster ✓	**F.** A group of data points bunched closely together.

1. *Association*

As study time increases, test scores increase. Positive or negative association?

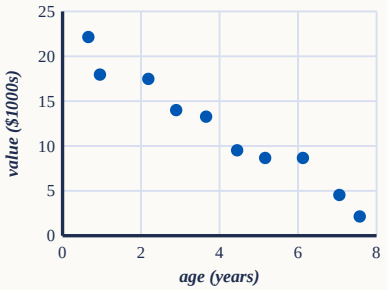
2. *Association*

As a car’s age increases, its value decreases. Positive or negative?



Answer:

answer: positive

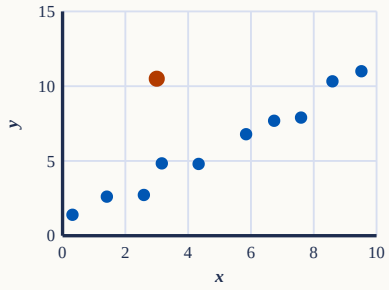


Answer:

answer: negative

3. *Vocabulary*

A point lying far from the pattern is called a(n) ___.

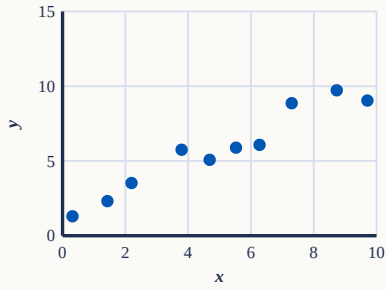


Answer:

answer: outlier

4. *Vocabulary*

A graph of paired data points is called a ___.

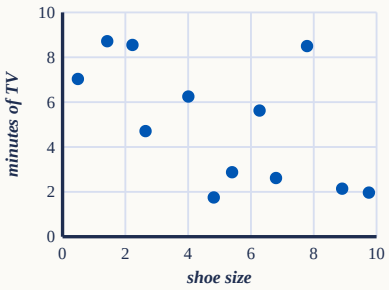


Answer:

answer: scatterplot

5. *Association*

If x and y show no pattern at all, there is ___ association.

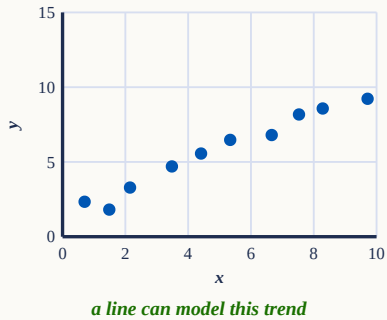


Answer:

answer: no

6. *Vocabulary*

A line drawn to model the trend of a scatterplot is the line of ___.

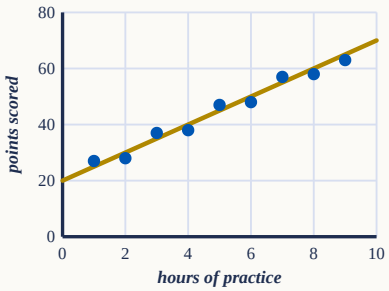


Answer:

answer: best fit

7. *Line of best fit*

Does a line of best fit have to go through every point? (yes or no)



Answer:

answer: no

SCORE 4.0 Target Content

NC.8.SP.2, NC.8.SP.3, NC.8.SP.4 Use a line of best fit to predict, and read two-way tables.

• *I can use a line of best fit to make predictions and read values from a two-way table.*

Use the line of best fit and the two-way table to answer each question.

Two-way table — use for the questions below

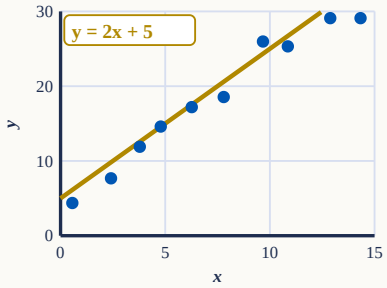
	Likes Pizza	Does Not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

8. *Predict*

Line of best fit y = 2x + 5. Predict y when x = 10.

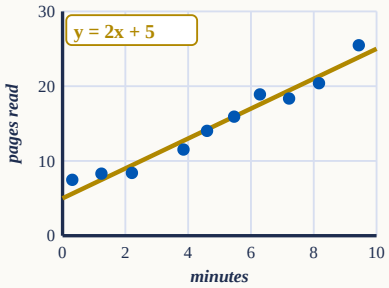
9. *Predict*

Line of best fit y = 2x + 5. Predict y when x = 0.



y = _____

answer: 25

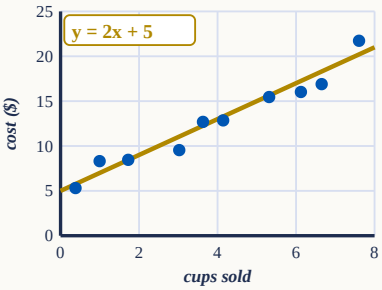


y = _____

answer: 5

10. *Interpret*

In y = 2x + 5, by how much does y increase for each increase of 1 in x?



y increases by _____

answer: 2

11. *Two-way table*

From the table, how many students like pizza in all?

students = _____

answer: 27

12. *Two-way table*

From the table, how many girls were surveyed?

girls = _____

answer: 20

13. *Two-way table*

From the table, how many students were surveyed in total?

students = _____

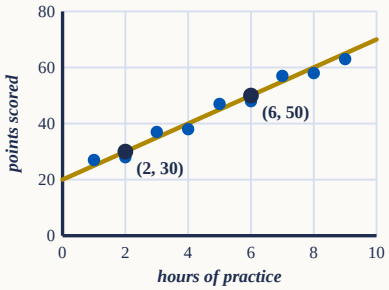
answer: 40

14. *Line of best fit*

The line of best fit goes through (2, 30) and (6, 50). Find its slope.

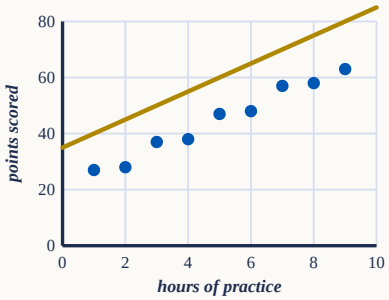
15. *Judge the fit*

Is this line a good fit for the data? (yes or no)



slope = _____

answer: 5



Answer: _____

answer: no

SCORE 5.0 Advanced Content

NC.8.SP.2 · NC.8.SP.4 Interpret relative frequencies and the meaning of slope in context.

• *I can find relative frequencies from a two-way table and interpret slope in a real situation.*

Use the two-way table above and each best-fit line to answer.

16. *Relative freq.*

From the table, what percent of boys like pizza?

percent = _____

answer: 60%

17. *Relative freq.*

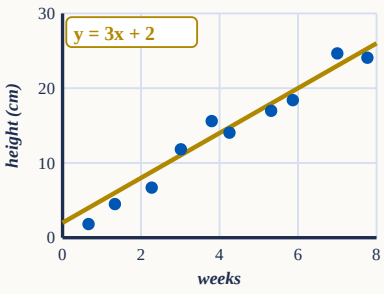
From the table, what percent of girls like pizza?

percent = _____

answer: 75%

18. *Predict*

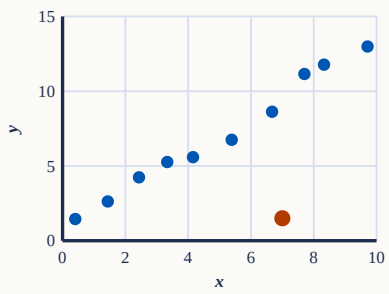
Best-fit line for plant height: y = 3x + 2. Predict the height at x = 6 weeks.



y = _____

19. *Vocabulary*

A scatterplot rises left to right, but one point sits far below the rest. That point is a(n) ___.



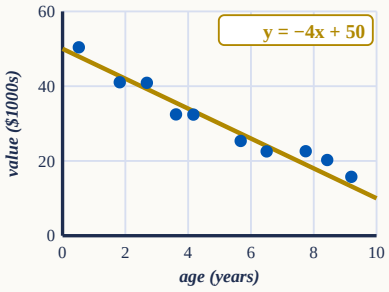
answer: 20

Answer:

answer: outlier

20. *Interpret*

Value vs. age: y = −4x + 50, where y is the value in $1,000s. How much does the value drop each year?

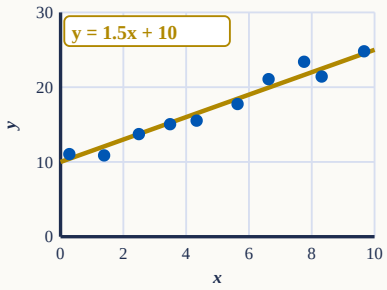


Answer:

answer: 4 thousand dollars ($4,000)

21. *Predict*

Line of best fit y = 1.5x + 10. Predict y when x = 8.

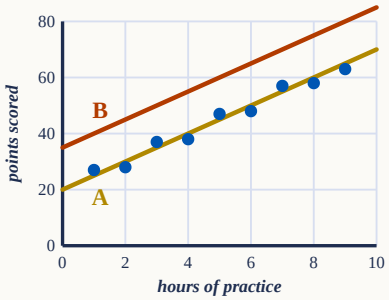


y =

answer: 22

22. *Compare two lines*

Line A and Line B are drawn on the scatter plot. Which line is the better fit, and why?



Answer:

answer: Line A. The points are close to it, with some above and some below.